Sketching Gradient Functions

Exam code: 9MA0

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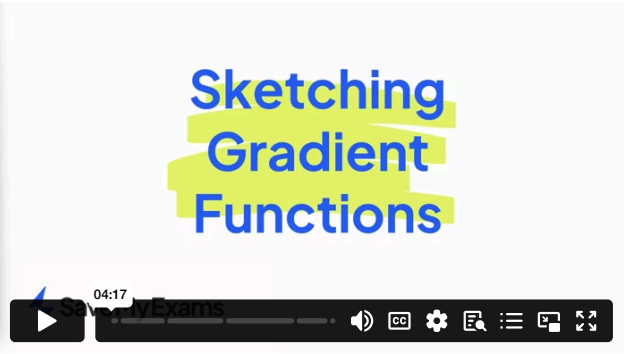

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Sketching gradient functions

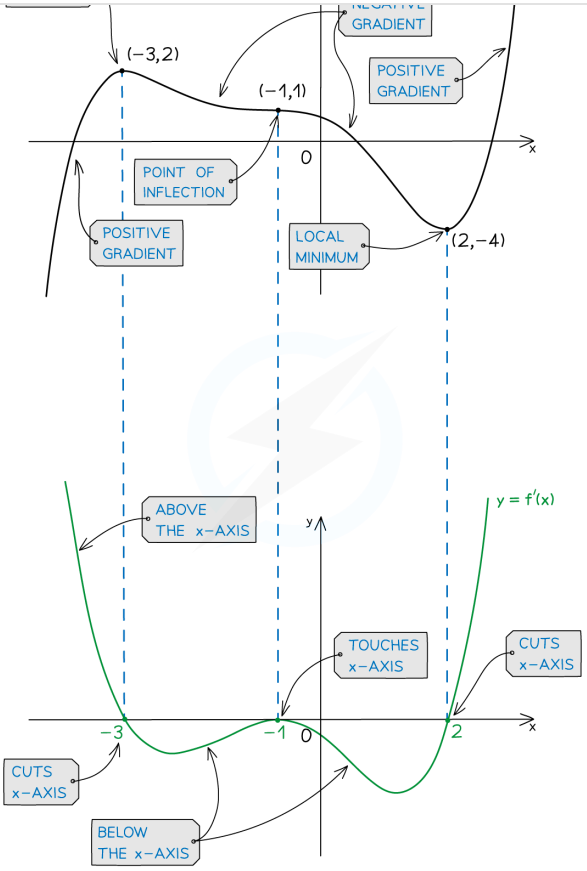
How do I sketch the gradient function from the original function?

- Using your knowledge of gradients and derivatives you can use the graph of a function to sketch the corresponding gradient function
- The behaviour of a function tells you about the behaviour of its gradient function

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



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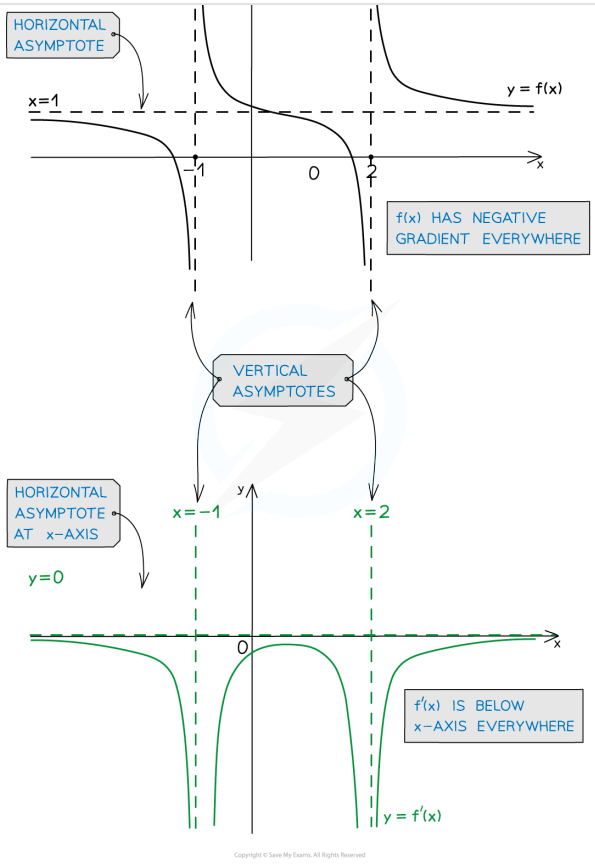
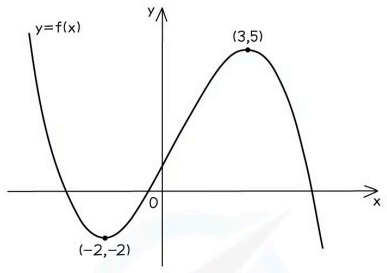


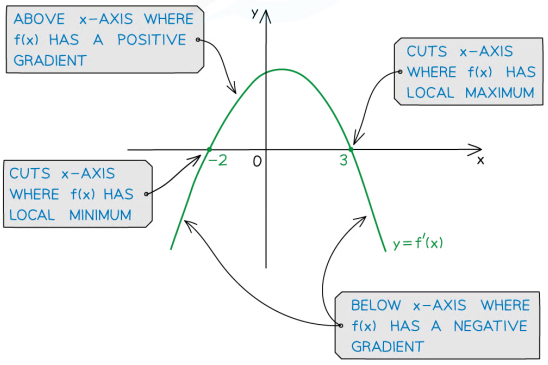
Examiner Tips and Tricks

- If **f(x)** is a smooth curve then **f'(x)** will also be a smooth curve.
- Take what you know about **f'(x)** (based on the table above) and then 'fill in the blanks' in between.
- If all you have is the graph of **f(x)** you will not be able to specify the coordinates of the *y*-intercept or any stationary points of **f'(x).**
- Be careful – points where **f(x)** cuts the *x*-axis don't tell you anything about the graph of **f'(x)**!

Worked Example



Sketch the graph of $y = f'(x)$.



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